

0 8 . 1

Give **three** properties of water that are important in biology.**[3 marks]**

1 _____

2 _____

3 _____

A student investigated the effect of different concentrations of sucrose solution on “chips” cut from a potato. Each chip had the same dimensions.

The student:

- weighed each chip at the start
- placed each chip in a separate test tube, each containing 10 cm³ of sucrose solution at a different concentration
- left the chips in the sucrose solution for 24 hours
- dried the surface of the chips and then weighed them again.

Table 4 shows the student’s results.

Table 4

Concentration of sucrose solution / mol dm ⁻³	Initial mass of chip / g	Final mass of chip / g	Ratio of final mass to initial mass of chips
0.0	2.79	3.82	
0.2	2.75	2.97	
0.4	2.78	2.67	
0.6	2.69	2.31	
0.8	2.72	2.20	
1.0	2.77	1.99	



0 8 . 2

The student produced the sucrose solutions with different concentrations from a concentrated sucrose solution.

Name the method she would have used to produce these sucrose solutions.

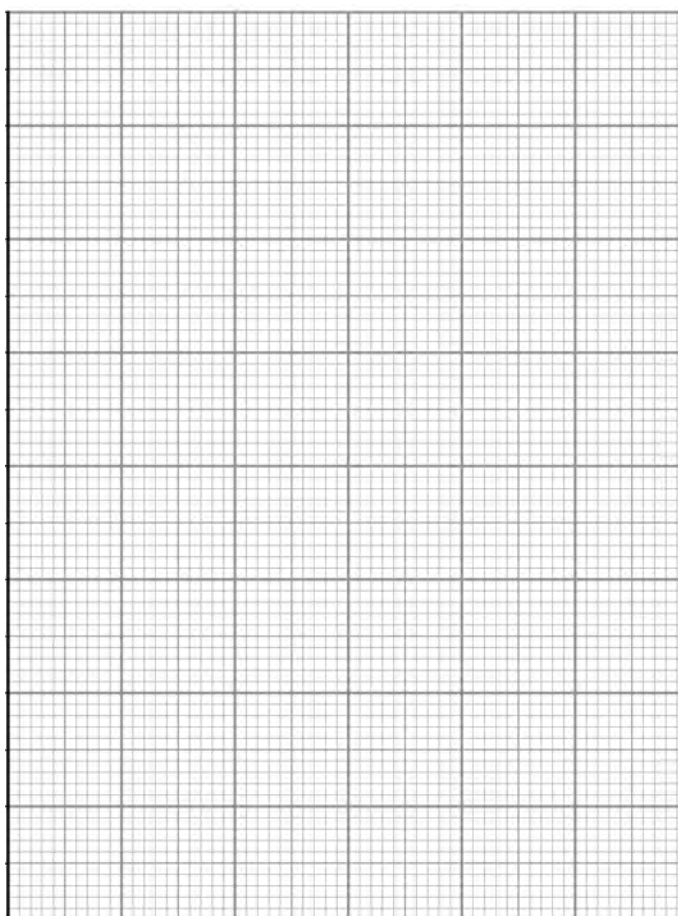
[1 mark]

Name of method _____

0 8 . 3

Calculate the ratio of final mass to initial mass of potato chips and plot a suitable graph of your processed data. Express the ratios in **Table 4** as a single number (for example 5.26:1 would be expressed as 5.26).

[3 marks]



Question 8 continues on the next page

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Explain the result for the chip in 0.8 mol dm^{-3} sucrose solution.

[2 marks]

9

